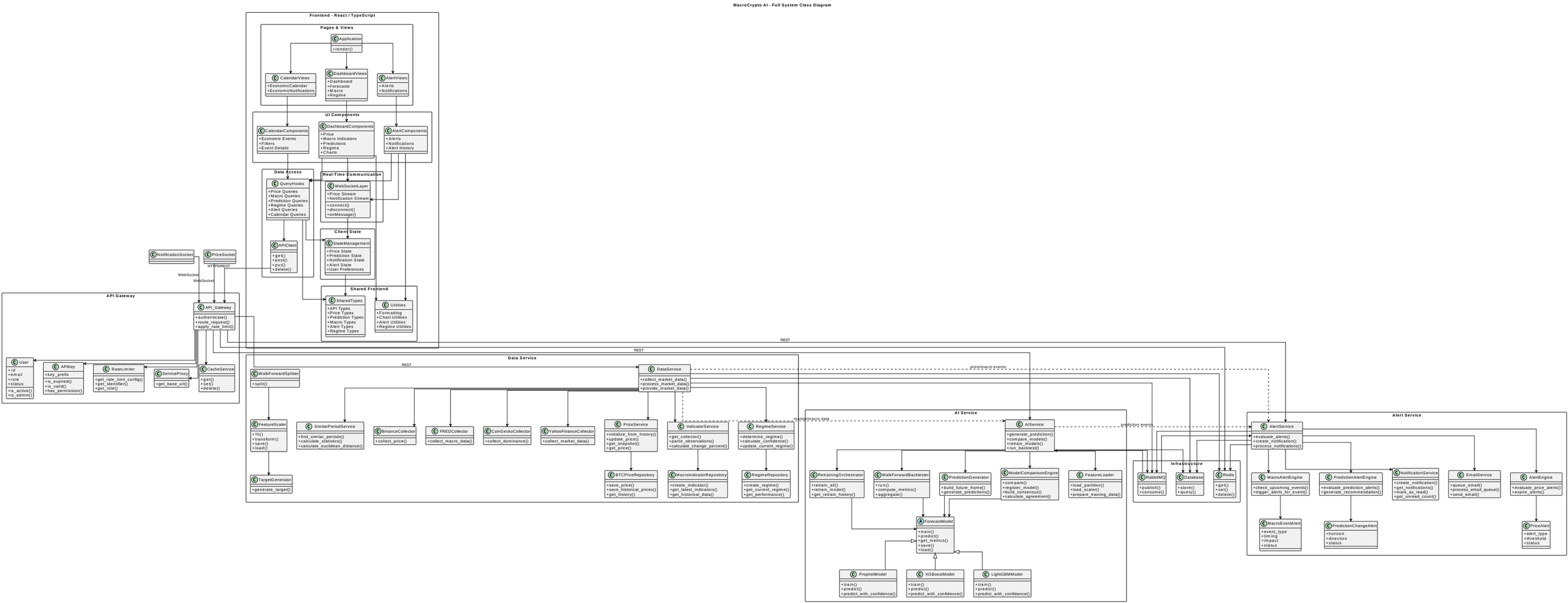

MacroCrypto AI - Full System Class Diagram
Frontend - React / TypeScript
Pages & Views
Application
+render()
CalendarViews
+EconomicCalendar
+EconomicNotifications
DashboardViews
+Dashboard
+Forecasts
+Macro
+Regime
AlertViews
+Alerts
+Notifications
UI Components
CalendarComponents
+Economic Events
+Filters
+Event Details
DashboardComponents
+Price
+Macro Indicators
+Predictions
+Regime
+Charts
AlertComponents
+Alerts
+Notifications
+Alert History
Data Access
QueryHooks
+Price Queries
+Macro Queries
+Prediction Queries
+Regime Queries
+Alert Queries
+Calendar Queries
Real-Time Communication
WebSocketLayer
+Price Stream
+Notification Stream
+connect()
+disconnect()
+onMessage()
Client State
StateManagement
+Price State
+Prediction State
+Notification State
+Alert State
+User Preferences
APIClient
+get()
+post()
+put()
+delete()
NotificationSocket
PriceSocket
HTTP/REST
WebSocket
WebSocket
Shared Frontend
SharedTypes
+API Types
+Price Types
+Prediction Types
+Macro Types
+Alert Types
+Regime Types
Utilities
+Formatting
+Chart Utilities
+Alert Utilities
+Regime Utilities
API Gateway
API_Gateway
+authenticate()
+route_request()
+apply_rate_limit()
User
+id
+email
+role
+status
+is_active()
+is_admin()
APIKey
+key_prefix
+is_expired()
+is_valid()
+has_permission()
RateLimiter
+get_rate_limit_config()
+get_identifier()
+get_role()
ServiceProxy
+get_base_url()
CacheService
+get()
+set()
+delete()
REST
REST
REST
Data Service
WalkForwardSplitter
+split()
DataService
+collect_market_data()
+process_market_data()
+provide_market_data()
price/macro events
FeatureScaler
+fit()
+transform()
+save()
+load()
SimilarPeriodService
+find_similar_periods()
+calculate_statistics()
+calculate_euclidean_distance()
BinanceCollector
+collect_price()
FREDCollector
+collect_macro_data()
CoinGeckoCollector
+collect_dominance()
YahooFinanceCollector
+collect_market_data()
PriceService
+initialize_from_history()
+update_price()
+get_snapshot()
+get_price()
IndicatorService
+get_collector()
+parse_observations()
+calculate_change_percent()
RegimeService
+determine_regime()
+calculate_confidences()
+update_current_regime()
TargetGenerator
+generate_target()
BTCPriceRepository
+save_price()
+save_historical_prices()
+get_history()
MacroIndicatorRepository
+create_indicator()
+get_latest_indicators()
+get_historical_data()
RegimeRepository
+create_regime()
+get_current_regime()
+get_performance()
AI Service
AIService
+generate_prediction()
+compare_models()
+retrain_models()
+run_backtest()
price/macro data
prediction events
RetrainingOrchestrator
+retrain_all()
+retrain_model()
+get_retrain_history()
WalkForwardBacktester
+run()
+compute_metrics()
+aggregate()
PredictionGenerator
+build_future_frame()
+generate_predictions()
ModelComparisonEngine
+compare()
+register_model()
+build_consensus()
+calculate_agreement()
FeatureLoader
+load_partition()
+load_scaler()
+prepare_training_data()
ForecastModel
+train()
+predict()
+get_metrics()
+save()
+load()
ProphetModel
+train()
+predict()
+predict_with_confidence()
XGBoostModel
+train()
+predict()
+predict_with_confidence()
LightGBMModel
+train()
+predict()
+predict_with_confidence()
Infrastructure
RabbitMQ
+publish()
+consume()
Database
+store()
+query()
Redis
+get()
+set()
+delete()
Alert Service
AlertService
+evaluate_alerts()
+create_notification()
+process_notifications()
MacroAlertEngine
+check_upcoming_events()
+trigger_alerts_for_event()
PredictionAlertEngine
+evaluate_prediction_alerts()
+generate_recommendation()
NotificationService
+create_notification()
+get_notifications()
+mark_as_read()
+get_unread_count()
EmailService
+queue_email()
+process_email_queue()
+send_email()
AlertEngine
+evaluate_price_alerts()
+expire_alerts()
MacroEventAlert
+event_type
+timing
+impact
+status
PredictionChangeAlert
+horizon
+direction
+status
PriceAlert
+alert_type
+threshold
+status